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COURSE INFORMATION

Learning Outcomes

- **LO1** describe the basic concepts of descriptive and inferential statistics.
- **LO2** calculate the statistical measurement which related to descriptive and inferential statistics.
- **LO3** Use Microsoft Excel to do data analysis.

Assessment Method

Teaching and Learning Strategies

- Exercise and solve problem with students
- Hands-on Practice
- Individual and Team Assignment
- Individual Exercises
- Lecture
- Software Exploring

Evaluation Weight

CLASS TYPE	COMPONENT	WEIGHT
LEC	THEORY: ASSIGNMENT	30%
	THEORY: FINAL EXAM	35%
	THEORY: MID EXAM	35%

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RULES IN CLASS

- **DON'T FORGET TO SCAN "ROOM CHECK-IN"**
- **Tepat waktu 07.20 – Toleransi keterlambatan 10 menit – (lebih-absence)**
- **Quiz akan dilaksanakan random day 15 menit pertama**
- **2 kali absence tidak bisa mengikuti final exam**
- **Tidak diperbolehkan makan dan minum**
- **Tidak boleh ribut satu sama lain**
- **Plagiarsm dan menyontek = DROP OUT**

Pay attention on GSLC Session

Guided Self Learning Class

- **Session 4 and 9**

- Read carefully the assignment on the forum bimay
- Pay attention on the deadline submission
- You must give feedback by giving comment on the forum. If not – absence.

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Komti / Head Prefect - LA95 STAT6202051

- **Volunteer**

???

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Course : STAT6202051 – Basic Statistics

Defining and Collecting Data

Session 1



DEFINING VARIABLES

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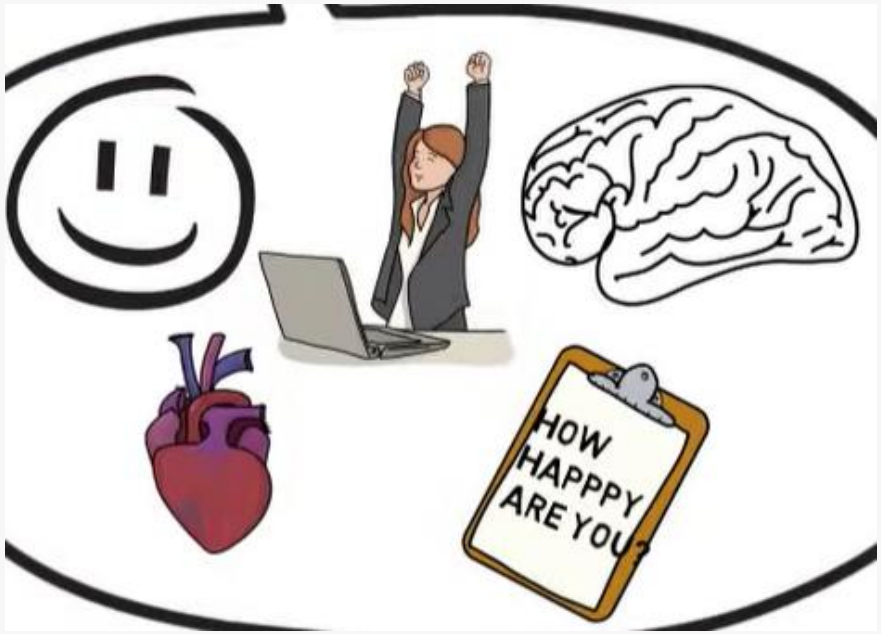
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Defining Variables

- When a proper business problem or objective has been identified, you can begin to define your data.
- You define data by defining variables
- You assign an **operational definition** to each variables you identify and specify the type of variable and the scale, or type of measurement, the variable uses

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Determine how we measure "HAPPINESS" ?



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Example case 1

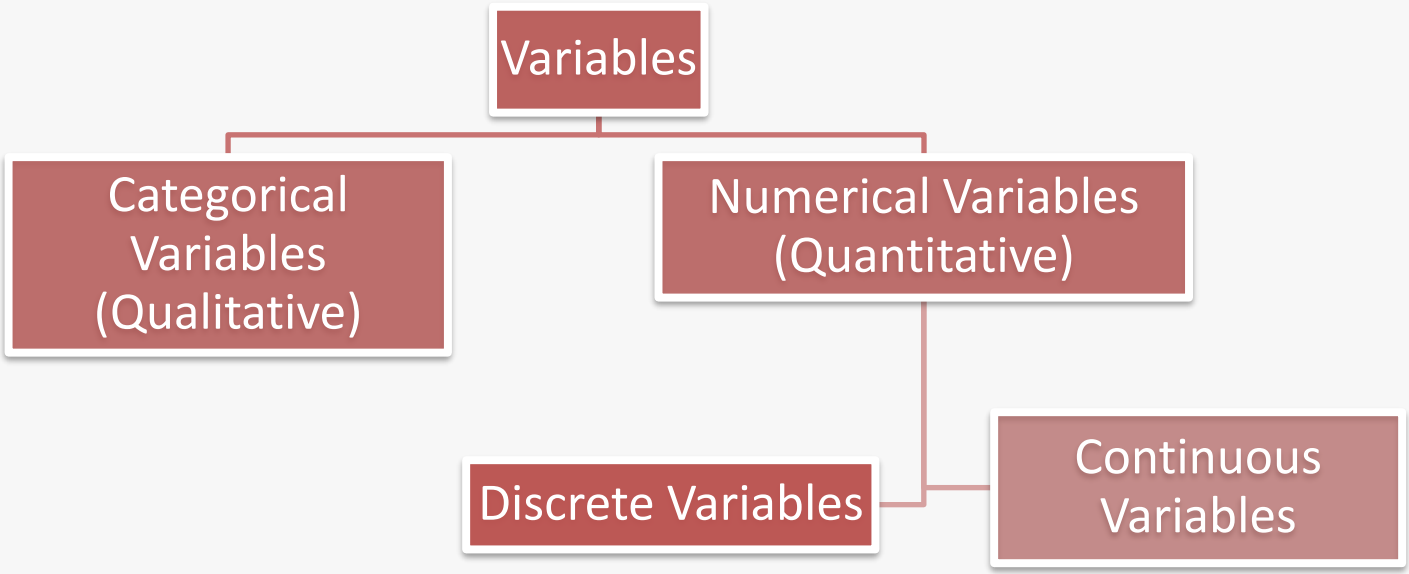
You been hired by Good Tunes & More (GT&M), a local electronics retailer, to assist in establishing a fair and reasonable price for Whitney Wireless, a privately-held chain that GT&M seeks to acquire. You need data that would help to analyse and verify the contents of the wireless company's basic financial statements. A GT&M manager suggests that one variable you should use is monthly sales. What do you do?

Solution

- Having first confirmed with the GT&M financial team that monthly sales is a relevant variable of interest; you develop an operational definition for this variable.
- Does this variable refer to sales per month for the entire chain or for individual stores?
- Does the variable refer to net or gross sales?
- Do the monthly sales data represent number of units sold or currency amounts?
- If the data are currency amounts, are they expressed in U.S. dollars?
- After getting answers to these and similar questions, you draft an operational definition for ratification by others working on this project.

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Classifying Variables by Type



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Classifying Variables by Type

You need to know the type of data that a variable defines in order to choose statistical methods that are appropriate for that data. Broadly, all variables are either **numerical**, variables whose data represent a counted or measured quantity, or **categorical**, variables whose data represent categories

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Classifying Variables by Type

- **Discrete** numerical variables have data that arise from a counting process. Discrete numerical variables include variables that represent a "number of something," such as the monthly number of smartphones sold in an electronics store.
- **Continuous** numerical variables have data that arise from a measuring process. The variable "the time spent waiting on a checkout line" is a continuous numerical variable because its data represent timing measurements.

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Measurement and Scales

You identify the **measurement scale** that the data for a variable represent, as part of defining a variable. The measurement scale defines the ordering of values and determines if differences among pairs of values for a variable are equivalent and whether you can express one value in terms of another.

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You define numerical variables as using either an **interval scale**, which expresses a difference between measurements that do not include a true zero point, or a **ratio scale**, an ordered scale that includes a true zero point.

Categorical variables use measurement scales that provide less insight into the values for the variable. For data measured on a **nominal scale**, category values express no order or ranking. For data measured on an **ordinal scale**, an ordering or ranking of category values is implied.

Data	Scale, Type	Values
Cellular provider	nominal, categorical	AT&T, T-Mobile, Verizon, Other, None
Excel skills	ordinal, categorical	novice, intermediate, expert
Temperature (°F)		–459.67°F or higher
SAT Math score		a value between 200 and 800, inclusive
Item cost (in $)		$0.00 or higher

APPLYING THE CONCEPT

1. U.S. businesses are listed by size: small, medium, and large. Explain why business size is an example of an ordinal-scaled variable.

2. For each of the following variables, determine whether the variable is categorical or numerical and determine its measurement scale. If the variable is numerical, determine whether the variable is discrete or continuous

a. Number of cellphones in the household

b. Monthly data usage (in MB)

c. Number of text messages exchanged per month

d. Voice usage per month (in minutes)

e. Whether the cellphone is used for email

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Solution

1. Small, medium, and large sizes imply order but do not specify how much the size of the business increases at each level

2. **(a)** The number of cellphones is a numerical variable that is discrete because the outcome is a count. It is ratio scaled because it has a true zero point.
(b) Monthly data usage is a numerical variable that is continuous because any value within a range of values can occur. It is ratio scaled because it has a true zero point.
(c) Number of text messages exchanged per month is a numerical variable that is discrete because the outcome is a count. It is ratio scaled because it has a true zero point.
(d) Voice usage per month is a numerical variable that is continuous because any value within a range of values can occur. It is ratio scaled because it has a true zero point.
(e) Whether a cellphone is used for email is a categorical variable because the answer can be only yes or no. This also makes it a nominal-scaled variable.

COLLECTING DATA

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- A **population** contains all the items or individuals of interest that you seek to study. All of the GT&M sales transactions for a specific year, all of the full-time students enrolled in a college, and all of the registered voters in Ohio are examples of populations.

- A **sample** contains only a portion of a population of interest. You analyze a sample to estimate characteristics of an entire population. You might select a sample of 200 GT&M sales transactions, a sample of 50 full-time students selected for a marketing study, or a sample of 500 registered voters in Ohio in lieu of analyzing the populations identified in this paragraph.

DATA SOURCES

Data sources arise from the following activities:

- Capturing data generated by ongoing business activities
- Distributing data compiled by an organization or individual
- Compiling the responses from a survey
- Conducting a designed experiment and recording the outcomes of the experiment
- Conducting an observational study and recording the results of the study

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APPLYING THE CONCEPT

The American Community Survey (www.census.gov/acs) provides data every year about communities in the United States. Addresses are randomly selected, and respondents are required to supply answers to a series of questions.

a. Which of the sources of data best describe the American Community Survey?

b. Is the American Community Survey based on a sample or a population?

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Solution

(a) Data distributed by an organization or individual

(b) Sample.

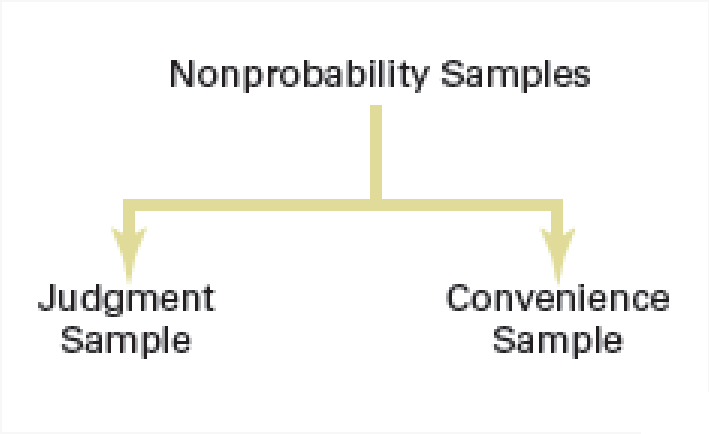
TYPES OF SAMPLING METHODS

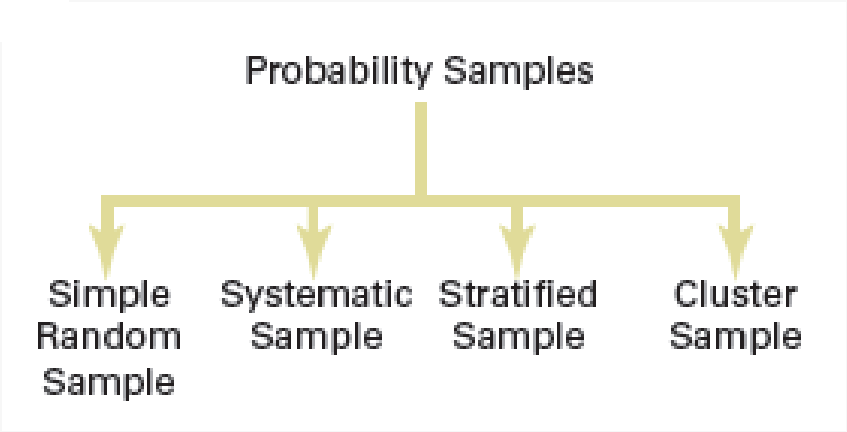
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Probability Sample	Nonprobability Sample

- In a **nonprobability sample**, you select the items or individuals without knowing their probabilities of selection.
- In a **probability sample**, you select items based on known probabilities. Whenever possible, you should use a probability sample as such a sample will allow you to make inferences about the population being analyzed.

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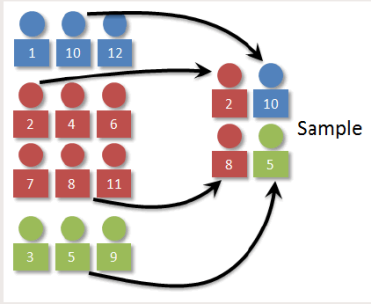


Non probability sampling

- In a **judgment sample**, you collect the opinions of preselected experts in the subject matter. Although the experts may be well informed, you cannot generalize their results to the population
- In a **convenience sample**, you select items that are easy, inexpensive, or convenient to sample. For example, in a warehouse of stacked items, selecting only the items located on the tops of each stack and within easy reach would create a convenience sample.

Probability sampling

- Stratified sampling
  - Divide the lot into subgroups (palet, box, location)
  - Randomly inspect units from within each subgroup
  - May have the same probability for each unit to end up in the sample
  - Samples are not necessarily equally likely



- Systematic sampling
  - Randomly pick a starting point
  - Pick every k-th product in the lot



- Cluster sampling
  - Pick one or more representative subgroup(s)
  - Inspect all units from the subgroup(s)



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Example

- Interview every 10th student who enters the school in the morning
  a) Simple Random
  b) Stratified
  c) Systematic
  d) Convenience

- Linda wanted to know students' opinion on the new schedule at school. She survey's the first 30 students who come into her class.
  a) Convenience Sampling
  b) Systematic Random Sampling
  c) Simple Random Sampling
  d) Stratified

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TYPES OF SURVEY ERRORS

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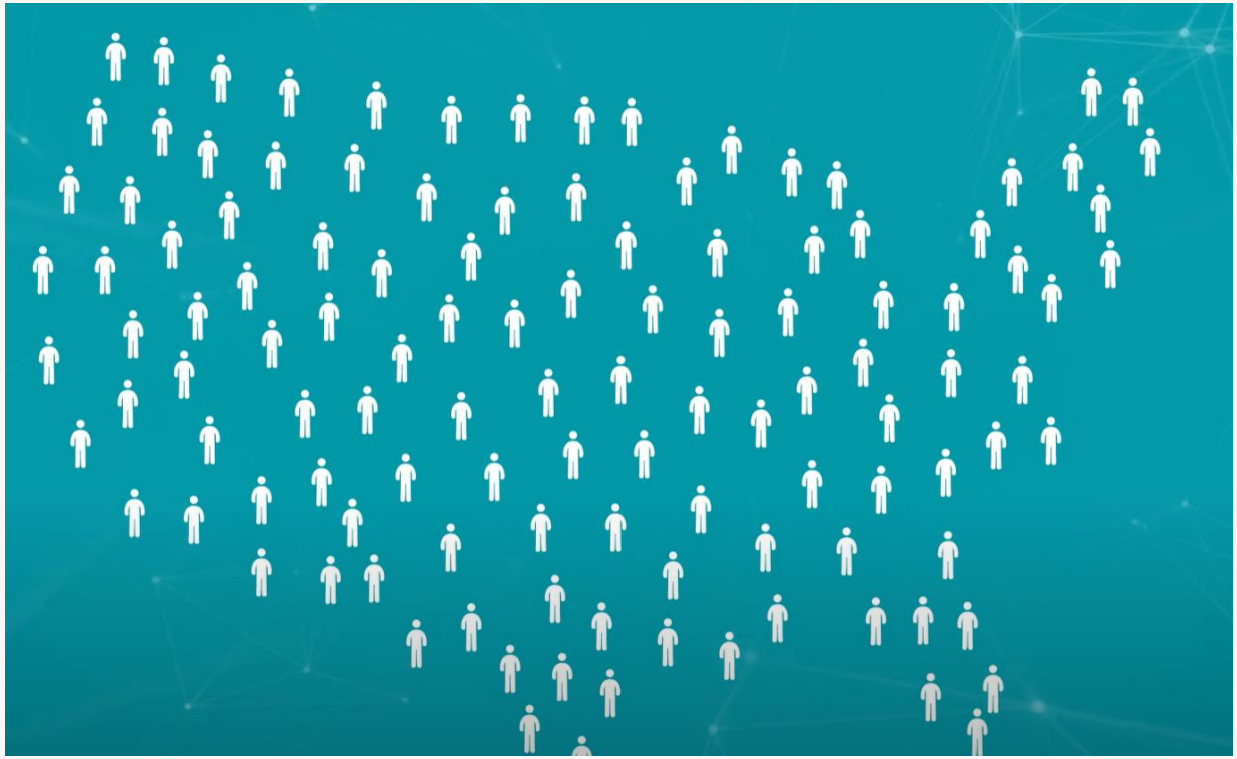
When you collect data using the compiled responses from a survey, you must verify two things about the survey in order to make sure you have results that can be used in a decision-making process. You must evaluate the validity of the survey to make sure the survey does not lack objectivity or credibility. To do this, you evaluate the purpose of the survey, the reason the survey was conducted, and for whom the survey was conducted.

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Coverage error occurs if certain groups of items are excluded from the frame so that they have no chance of being selected in the sample or if items are included from outside the frame.

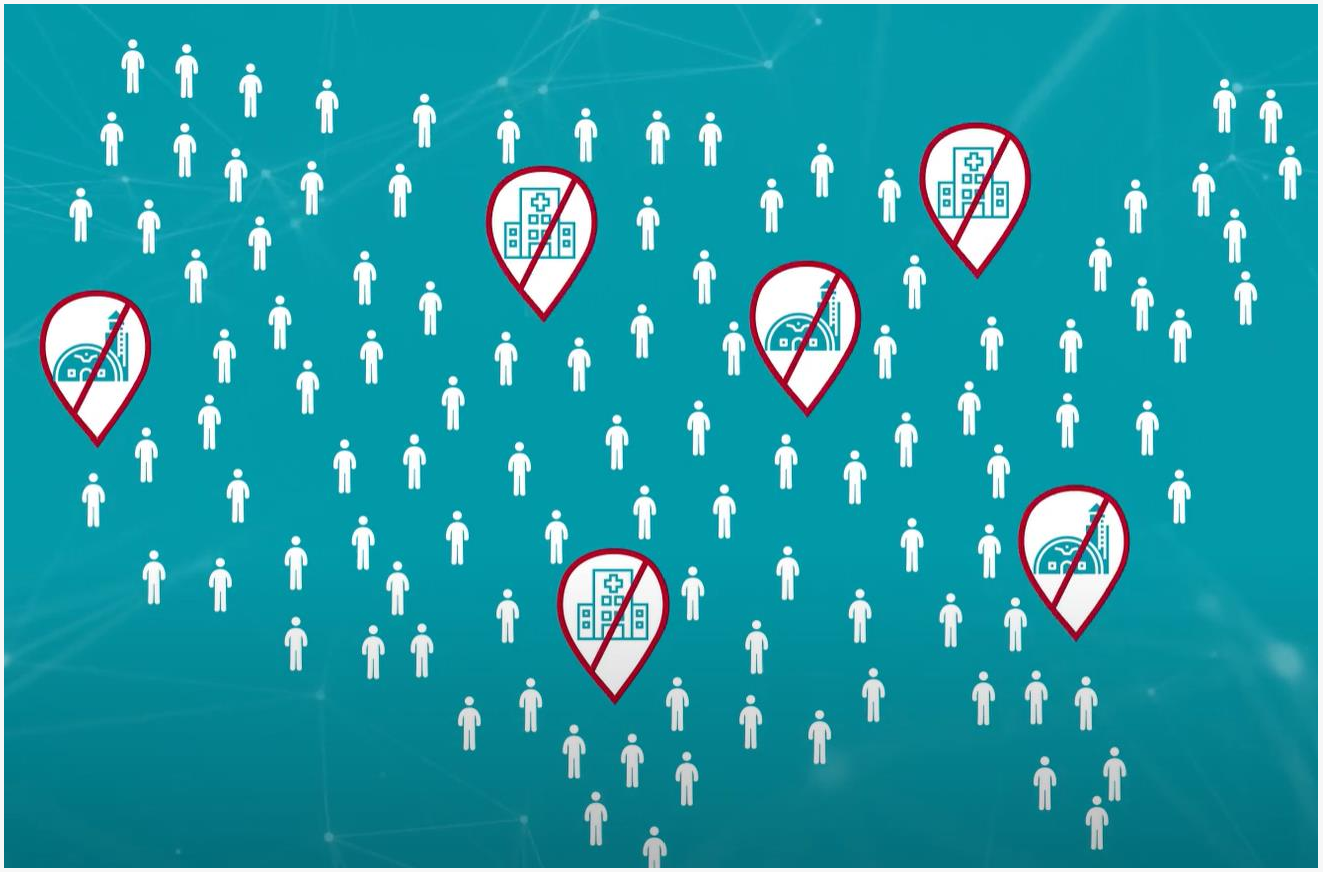
Coverage error results in a **selection bias**. If the frame is inadequate because certain groups of items in the population were not properly included, any probability sample selected will provide only an estimate of the characteristics of the frame, not the *actual* population.

Coverage error examples (bias)



Adults living in Indonesia

Coverage error examples (bias)



But the survey wants does not want to include the military, hospital and institution.

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Not everyone is willing to respond to a survey. **Nonresponse error** arises from failure to collect data on all items in the sample and results in a **nonresponse bias**.

Because you cannot always assume that persons who do not respond to surveys are similar to those who do, you need to follow up on the nonresponses after a specified period of time. You should make several attempts to convince such individuals to complete the survey and possibly offer an incentive to participate.

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The follow-up responses are then compared to the initial responses in order to make valid inferences from the survey. The mode of response you use, such as face-to-face interview, telephone interview, paper questionnaire, or computerized questionnaire, affects the rate of response. Personal interviews and telephone interviews usually produce a higher response rate than do mail surveys—but at a higher cost.

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Occurs when the are differences between the estimates from surveying a portion of the targeted population vs every unit in the population, like in a Census.

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In the practice of good survey research, you design surveys with the intention of gathering meaningful and accurate information. Unfortunately, the survey results you get are often only a proxy for the ones you really desire. Unlike height or weight, certain information about behaviours and psychological states is impossible or impractical to obtain directly.

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When surveys rely on self-reported information, the mode of data collection, the respondent to the survey, and or the survey itself can be possible sources of **measurement error**. Satisficing, social desirability, reading ability, and/or interviewer effects can be dependent on the mode of data collection.

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The social desirability bias or cognitive/memory limitations of a respondent can affect the results. And vague questions, double-barreled questions that ask about multiple issues but require a single response, or questions that ask the respondent to report something that occurs over time but fail to clearly define the extent of time about which the question asks (the reference period) are some of the survey flaws that can cause errors.

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Applying The Concepts

A survey indicates that the vast majority of college students own their own personal computers. What information would you prepare for this survey?

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Solution

Before accepting the results of a survey of college students, you might want to know, for example: Who funded the survey? Why was it conducted? What was the population from which the sample was selected? What sampling design was used? What mode of response was used: a personal interview, a telephone interview, or a mail survey? Were interviewers trained? Were survey questions field-tested? What questions were asked? Were the questions clear, accurate, unbiased, and valid? What operational definition of immediately and effortlessly was used? What was the response rate?

QUIZ

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https://www.menti.com/alg5uovntgdp



USING STATISTICS: DEFINING MOMENTS

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- Anda adalah manajer penjualan yang bertanggung jawab atas minuman terlaris di kategorinya. Selama bertahun-tahun, pesaing utama Anda telah memperoleh keuntungan penjualan, mengklaim produk yang rasanya lebih enak.
- Lebih buruk lagi, dengan cepat memperoleh pangsa pasar yang signifikan dengan mengorbankan produk Anda.
- Khawatir bahwa produk Anda akan segera kehilangan status nomor satu, Anda berusaha meningkatkan penjualan dengan meningkatkan cita rasa produk. Anda bereksperimen dan mengembangkan formulasi minuman baru.
- Dengan menggunakan metode yang diajarkan dalam sesi ini, Anda melakukan survei dan menemukan bahwa orang-orang sangat menyukai formulasi yang lebih baru, dan Anda memutuskan untuk menggunakan formulasi baru itu ke depan, setelah secara statistik menunjukkan bahwa orang lebih menyukai formulasi rasa yang baru.
- **Apa yang bisa salah dari masalah di atas?**

Answer

- Menceritakan kembali kisah salah satu kesalahan pemasaran paling terkenal yang pernah ada, perubahan formulasi Coca-Cola di tahun 1980-an.
- Dalam hal ini, manajer merek Coke begitu fokus pada rasa Pepsi dan saudara kandung yang baru sukses, Diet Coke, sehingga mereka memutuskan hanya untuk menentukan variabel dan mengumpulkan data tentang minuman yang disukai pencicip dalam tes rasa buta.Ketika Coke Baru lebih disukai, bahkan daripada Pepsi, para manajer bergegas membuat formulasi baru ke dalam produksi.
- Dalam melakukannya, para manajer tersebut gagal untuk merefleksikan apakah hasil statistik tentang tes yang meminta orang untuk membandingkan satu ons sampel beberapa minuman akan menunjukkan apa pun tentang penjualan minuman.
- Lagi pula, orang-orang ditanya minuman mana yang rasanya lebih enak, bukan apakah mereka akan membeli minuman yang rasanya lebih enak itu di masa depan. Coke baru langsung gagal, dan manajer Coke membalikkan keputusan mereka hanya dalam 77 hari setelah memperkenalkan formulasi baru mereka

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ASSIGNMENT 1 →BIMAY

PAY ATTENTION
ON THE DEADLINE SUBMISSION

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Following Session
Please bring laptop
For using Excel Data Analysis

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David M. Levine, David F. Stephan, Kathryn A. Szabat. Statistics for Managers using Microsoft Excel. Pearson, 8th Edition.

Thank You
Kumara Dharaka, ST. MSc.

JANGAN LUPA BAHAGIA